

Polynomials in Number Theory

Problem 1. Call a two-element subset of $\mathbb{N}$ *cute* if it contains exactly one prime number and one composite number. Determine all polynomials $f \in \mathbb{Z}[x]$ such that for every cute subset $\{p, q\}$, the subset $\{f(p) + q, f(q) + p\}$ is cute as well.

Problem 2. Find all polynomials $P(x)$ with integer coefficients such that $P(P(n) + n)$ is a prime number for infinitely many integers n .

Problem 3. Determine all nonempty finite sets of positive integers $\{a_1, \dots, a_n\}$ such that $a_1 \cdots a_n$ divides $(x + a_1) \cdots (x + a_n)$ for every positive integer x .

Problem 4. Big Bird has a polynomial P with integer coefficients such that n divides $P(2^n)$ for every positive integer n . Prove that Big Bird's polynomial must be the zero polynomial.

Problem 5. Find all polynomials W with integer coefficients satisfying the following condition: For every natural number n , $2^n - 1$ is divisible by $W(n)$.

Problem 6. Find all polynomials $P(x)$ with integer coefficients such that $P(n)$ divides $n^{n-1} - 1$ for all sufficiently large positive integer n .

Problem 7. Determine all polynomials f with integer coefficients such that $f(p)$ is a divisor of $2^p - 2$ for every odd prime p .

Problem 8. Let $f(x), g(x)$ be two polynomials with integer coefficients. It is known that for infinitely many primes p , there exists an integer m_p such that

$$f(a) \equiv g(a + m_p) \pmod{p}$$

holds for all $a \in \mathbb{Z}$. Prove that there exists a rational number r such that

$$f(x) = g(x + r).$$